

*exp case 2.sci

```
1 clc;
2 clear;
3 r = 3;
4 phi = 60;
5 z = -5;
6
7 x = r * cosd(phi);
8 y = r * sind(phi);
9
10 disp("Cylindrical to Cartesian coordinate system P(x,y,z) =");
11 disp([x y z]);
12
13 R = sqrt(r^2 + z^2);
14 theta = acosd(z / R);
15
16 disp("Cylindrical to Spherical coordinate system S(R,phi,theta) =");
17 disp([R phi theta]);
18
```

Scilab 2025.1.0 Console

File Edit Control Applications ?



Scilab 2025.1.0 Console

```
"Cylindrical to Cartesian coordinate system P(x,y,z) ="
1.5  2.5980762  -5.
"Cylindrical to Spherical coordinate system S(R,phi,theta) ="
5.8309519  60.  149.03624
```

--> |

TEST Case - 2

$$x = +3$$

$$\phi = 60^\circ$$

$$z = -5$$

$$x = \rho \cos \phi = 5 \cos(60^\circ) = \cancel{2.5} 1.5$$

$$y = \rho \sin \phi = 5 \sin(60^\circ) = \cancel{4.33} 2.5981$$

$$z = -5 =$$

$$\rho = \sqrt{x^2 + z^2} = \sqrt{3^2 + (-5)^2} = 5.83$$

$$\theta = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{-5}{5.83}\right) = \cancel{30.948} 149.03$$

$$\phi = \phi = 60^\circ$$

Cartesian (1.5, 2.5981, -5)

Spherical (5.831, 60°, 149.036°)

O/p X m